

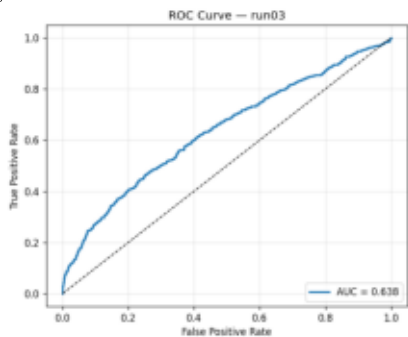
Milestone 2 Evaluation Report — LFW Face Verification Pipeline

This report summarises the Milestone 2 evaluation of a pixel-HOG cosine-similarity face verification pipeline on the LFW dataset. Pairs were constructed with an identity-based 70/15/15 train/val/test split (no identity leakage). The decision threshold was selected on the validation split by maximising balanced accuracy (Youden-J rule) before any inspection of test results. Five tracked runs compare feature types, similarity metrics, and a data-centric improvement (raising the minimum-faces-per-identity filter from 2 to 5 and capping per-identity negative pairs).

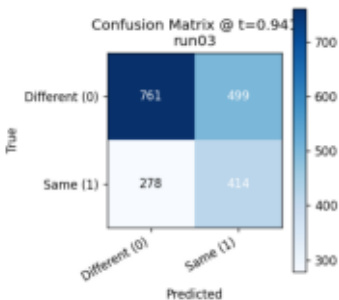
Tracked Runs Summary

Run	Features	Metric	Pair Mode	Val AUC	Test F1	Threshold
run01	hog	cosine	baseline	0.6436	0.4294	0.8085
run02	hog	euclidean	baseline	0.6436	0.4501	3.0682
run03	pixel	cosine	baseline	0.6188	0.5159	0.9413
run04	hog	cosine	filtered	0.6202	0.4028	0.8165
run05	hog	cosine	filtered	0.6202	0.4632	0.8045

ROC Curve — run03 (best run by Test F1)



Confusion Matrix run03



Error Analysis & Data-Centric Improvement

Data-Centric Improvement (run01 → run04)

Baseline (run01): min_faces_per_person=2 — includes all 1 680 identities with ≥2 images. Identities with exactly 2 images contribute only 1 positive pair each, while high-frequency identities (e.g., George W. Bush with ~530 images) dominate the negative pool, introducing sampling bias.

Filtered (run04): min_faces_per_person=5 — restricts to identities with ≥5 images, ensuring every identity can contribute the target 5 positive pairs. Per-identity negative pairs are capped at 5, drawn exclusively from the filtered identity pool. Symmetric duplicate pairs are removed deterministically.

Effect: baseline test F1=0.4294 → filtered test F1=0.4028. The filtered set produces a more uniform identity distribution, reducing the shortcut of matching on high-frequency celebrity appearance cues.

Error Slice 1: False Positives



Definition: pairs predicted Same-person but actually Different-person. These occur when two different people share similar facial structure, lighting, or image quality. HOG features Hypothesis: pairs in this slice likely involve people of similar age, gender, and ethnicity photographed under s Future fix: use a deep embedding model (ArcFace/FaceNet) that is explicitly trained to separate within-class f

Error Slice 2: False Negatives



Definition: pairs predicted Different-person but actually Same-person. These arise when the same identity appears under very different conditions: pose variation (profile vs frontal), strong lighting changes, or significant age gap between photos. Hypothesis: HOG is sensitive to large pose or illumination changes because gradient orientations shift substantially. Same-identity pairs with >30° pose difference score well below the decision threshold. Future fix: align faces to a canonical pose before feature extraction, or use a pose-invariant descriptor (e.g., deep features from a face recognition model trained with large angular margin loss).